



Handheld keypad and mouse?

Want a compact keyboard and mouse functionality? Try out the Portronics mini wireless keyboard which comes with a trackpad to be used as a mouse pointer

A computainment solution for your HDTV

Get the HD back in your HDTVs and also do basic computing tasks. This DIY shows you how to get a sleek system to fulfill this.

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You're bound to find a lot more High Definition TVs nowadays than traditional CRTs, thanks to the falling prices of LCD panels. The only way a majority of the users enjoy HD content on their screens is if they have an HD set-top box. But what about the collection of HD movies that you have on your system's hard drive? Streaming it is one option, but the control lies with the system which will most probably be in some other room. The other option is to connect your laptop to your HDTV via an HDMI cable and playback your movie. Honestly, how many times can you do that? What if you could have a complete entertainment box around your HDTV which can also allow basic computing?

Welcome to the world of home theatre PCs. The main purpose behind this DIY was to have a decent rig in the most compact form factor, which apart from playing HD content also allows you to surf the web, work on your documents and allow very basic gaming on your flat screen TV. We're assuming that you have a HDTV and/or speaker system.

What should you expect from an entry-level HTPC rig? Well for starters, it should play all your HD content flawlessly and we mean 1080p onboard. If the rig tends to drop frames while rendering or the output is jerky, then the purpose is lost. Upscaling SD content for viewing on 32-inches and above should be quite good. Online streamed video should be handled with ease.

Another critical aspect pertaining to building a sleek HTPC is heat and noise management. You surely don't want a system that heats up and makes the

system unstable. Since the system will be used for viewing HD movies, you want the rig to not make any noise, be it from the power supply or the CPU fan. You want a quiet system.

Keeping these points in mind, you should be very careful while selecting the components.

Test scores

Now there are various ways to go about building an HTPC. With every major motherboard manufacturer coming out with Mini ITX form factor boards (we looked at this form factor as we wanted a compact system that could be placed beside the TV). We decided to use compo-

nents on both ends of the spectrum.

For the high-end HTPC system we used an Intel Core i5-2500 processor with a clock speed of 3.3 GHz and integrated Intel HD 3000 graphics and lodged it on a ZOTAC Z68 Mini ITX board with onboard Wi-Fi. This HTPC is more like a workhorse system, which can also be a compact desktop replacement provided you don't go for a discrete graphics card for gaming, else that will defeat the purpose of building an HTPC. Plus, with an Intel Sandy Bridge processor's integrated graphics, you can watch all your HD content flawlessly and also transcode video files faster.

On the lower end of the segment we

How the rigs fared

Processor	Intel Core i5-2500	AMD E350
Motherboard	ZOTAC Z68ITX A-E	ECS HDC I2
RAM	2x2GB Corsair Dominator DDR3	2x2GB Corsair Dominator DDR3
HDD	300 GB Velociraptor	300 GB Velociraptor
Cabinet	Cooler Master CM Elite 100	Cooler Master CM Elite 100

PC Mark Vantage

PC Mark	9257	2549
Memories	4322	1499
TV and Movies	4796	1296
Gaming	6809	2075
Music	8398	2776
Communications	11011	2522
Productivity	8724	2495
HDD	3155	2791

3D Mark 2006

CPU	6478	783
Graphics	1593	859
Overall	4135	1950

Cinebench R10

Multi-core CPU	21692	2013
OpenGL	6740	1966
CPU Utilization during HD playback	<10 per cent	around 50
DivX Encoding (100MB VOB to DIVX)	20 secs	131 secs